3D Modelling Fundamentals

Computing and Mathematics

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Ollscoil Teicneolaíochta an Oirdheiscirt
South East Technological University

3D Modelling Fundamentals (A13461)

Short Title:	3D Modelling Fundamentals		
Faculty	Science and Computing		
Department	Computing and Mathematics		
Cluster	Graphic Design and Animation		
Author	PMCINERNEY		
Credits	5	**Level:**	Introductory

Description of Module / Aims

This module serves as an introduction to 3D modelling and comprises an emphasis on the modelling pipeline, from concept to model creation. Students will acquire a knowledge of the craft of 3D modelling and will gain experience in producing a number of 3D models using a number of essential tools within an industry-led 3D-creation toolset.

Programmes

`COMP-0595`	BSc (Hons) in Creative Computing (`WD_KCRCO_B`)	2 / 4 / M
`COMP-0595`	BSc in Multimedia Applications Development (`WD_KMULA_D`)	2 / 4 / M

Indicative Content

- Introduction to the 3D production pipeline, from concept to mesh creation
- Components of an industry-standard 3D modelling platform
- Polygon, subdivision, and NURB modelling
- Basic camera and viewing techniques
- Introductory shading and rendering

Learning Outcomes

On successful completion of this module, a student will be able to:

1. Describe the fundamentals of the 3D modelling workflow, from concept to a shaded mesh.
2. Identify and effectively navigate the main components and features of an industry standard 3D modelling production platform.
3. Compare various modelling approaches, i.e. polygon, sub-division and NURB-based modelling.
4. Demonstrate the use of appropriate tools for the creation and manipulation of 3D objects and object components.

Learning and Teaching Methods

- Interactive and open-forum lectures.
- Class discussions and presentations.
- Problem-based learning activities.
- Self-directed learning.

Assessment Methods

	Weighing	Outcomes Assessed
Continuous Assessment	100%	1,2,3,4
Assignment	30%	1
Project	70%	2,3,4

Assessment Criteria

<40%: Inability to identify and navigate the main components of a 3D production software platform. Inability to critically evaluate techniques used in the creation and exporting of 3D computer graphics models.

40%-49%: Ability to identify and navigate the main components of a 3D production software platform. Ability to describe key concepts of the 3D modelling pipeline.

50%-59%: Ability to discuss and critically evaluate key concepts of the 3D modelling pipeline and techniques.

60%-69%: Be able to solve problems within the framework of the 3D modelling pipeline by experimenting with appropriate skills and tools of the 3D production software.

70%-100%: All the above to an excellent level. Be able to analyse and design solutions to a high standard for a range of problems through the use and modification of appropriate skills and tools of the 3D production software.

Learning Modes

Learning Mode	F/T Hours	P/T Hours
Lecture	12	
Practical	36	
Independent Learning	87	

Supplementary Material

- Palamar, T. _Mastering Autodesk Maya 2016_. New York: Sybex, 2015.

Requested Resources

- COMPUTER LAB: BYOD Lab